

Aishwarya-eee

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6b9c00a · 2 weeks ago

82 lines (82 loc) · 49 KB

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In [1]:

```
import matplotlib.pyplot as plt
import pandas
from sklearn import tree
from sklearn.tree import DecisionTreeClassifier
import matplotlib.pyplot as plt
df = pandas.read_excel("/content/decision_tree.xlsx")
print(df)
d = {'UK': 0, 'USA': 1, 'N': 2}
df['Nationality'] = df['Nationality'].map(d)
d = {'YES': 1, 'NO': 0}
df['Go'] = df['Go'].map(d)
features = ['Age', 'Experience', 'Rank', 'Nationality']
X = df[features]
y = df['Go']
dtree = DecisionTreeClassifier()
dtree = dtree.fit(X, y)
tree.plot_tree(dtree,feature_names=features)
plt.show()
```

	Age	Experience	Rank	Nationality	Go
0	36	10	9	UK	NO
1	42	12	4	USA	NO
2	23	4	6	N	NO
3	52	4	4	USA	NO
4	43	21	8	USA	YES
5	44	14	5	UK	NO
6	66	3	7	N	YES
7	35	14	9	UK	YES
8	52	13	7	N	YES
9	35	5	9	N	YES
10	24	3	5	USA	NO
11	18	3	7	UK	YES
12	45	9	9	UK	YES

Rank <= 6.5
gini = 0.497
samples = 13
value = [6, 7]

True

gini = 0.0
samples = 5
value = [5, 0]

Age <= 35.5
gini = 0.375
samples = 4
value = [1, 3]

gini = 0.0
samples = 2
value = [0, 2]

gini = 0.0
samples = 1
value = [1, 0]

Age <= 40.5
gini = 0.5
samples = 2
value = [1, 1]

gini = 0.0
samples = 1
value = [0, 1]

False

Nationality <= 0.5
gini = 0.219
samples = 8
value = [1, 7]

gini = 0.0
samples = 4
value = [0, 4]

https://github.com/Aishwarya-eee/python-ML-Aishwarya/blob/main/_Decision_Tree_(9c).ipynb

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